

Homework Solutions: Differential and Difference Equations

GSE 510

Homework Problems: Submit individually

6. Consider the difference equation

$$x_{t+1} = 0.5x_t + 12.$$

(a) A steady state satisfies

$$x^* = 0.5x^* + 12.$$

Therefore,

$$x^* - 0.5x^* = 12,$$

so

$$0.5x^* = 12.$$

Hence,

$$x^* = 24.$$

(b) Suppose $x_0 = 0$. Then

$$x_1 = 0.5(0) + 12 = 12.$$

Next,

$$x_2 = 0.5(12) + 12 = 6 + 12 = 18.$$

Finally,

$$x_3 = 0.5(18) + 12 = 9 + 12 = 21.$$

(c) Yes, the sequence is converging to the steady state. Starting from

$$x_0 = 0,$$

we get

$$x_1 = 12, \quad x_2 = 18, \quad x_3 = 21.$$

These values are moving toward

$$x^* = 24.$$

(d) Since

$$x^* = 24,$$

subtract 24 from both sides of the equation:

$$x_{t+1} - 24 = 0.5x_t + 12 - 24.$$

Therefore,

$$x_{t+1} - 24 = 0.5x_t - 12.$$

Since

$$0.5(x_t - 24) = 0.5x_t - 12,$$

we can write

$$x_{t+1} - x^* = 0.5(x_t - x^*).$$

Thus,

$$a = 0.5.$$

- (e) The steady state is stable because

$$|a| = |0.5| < 1.$$

This means that each period, the distance between x_t and the steady state is multiplied by 0.5. Therefore, the sequence gets closer and closer to $x^* = 24$.

7. Consider the difference equation

$$y_{t+1} = 1.1y_t - 5.$$

- (a) A steady state satisfies

$$y^* = 1.1y^* - 5.$$

Rearranging,

$$y^* - 1.1y^* = -5.$$

Thus,

$$-0.1y^* = -5.$$

Therefore,

$$y^* = 50.$$

- (b) Suppose $y_0 = 60$. Then

$$y_1 = 1.1(60) - 5 = 66 - 5 = 61.$$

Next,

$$y_2 = 1.1(61) - 5 = 67.1 - 5 = 62.1.$$

Finally,

$$y_3 = 1.1(62.1) - 5 = 68.31 - 5 = 63.31.$$

- (c) The sequence moves away from the steady state. The steady state is

$$y^* = 50.$$

Starting from $y_0 = 60$, the sequence becomes

$$60, 61, 62.1, 63.31, \dots$$

which moves farther away from 50.

- (d) The steady state is unstable because the coefficient on y_t is

$$1.1,$$

and

$$|1.1| > 1.$$

- (e) The coefficient on y_t determines whether deviations from the steady state shrink or grow over time. Since the coefficient is greater than 1 in absolute value, deviations from the steady state become larger over time. That is why the steady state is unstable.

8. Consider the differential equation

$$\frac{dx}{dt} = 4x - 20.$$

- (a) A steady state satisfies

$$4x^* - 20 = 0.$$

Therefore,

$$4x^* = 20,$$

so

$$x^* = 5.$$

(b) If

$$x > x^* = 5,$$

then

$$4x - 20 > 0.$$

Therefore,

$$\frac{dx}{dt} > 0.$$

So x is increasing.

(c) If

$$x < x^* = 5,$$

then

$$4x - 20 < 0.$$

Therefore,

$$\frac{dx}{dt} < 0.$$

So x is decreasing.

- (d) The steady state is unstable. If x is slightly above 5, it increases and moves farther away from 5. If x is slightly below 5, it decreases and also moves farther away from 5. Thus, arrows point away from the steady state.

9. Suppose price adjusts according to

$$\frac{dp}{dt} = \gamma(D(p) - S(p)),$$

where

$$D(p) = 80 - p, \quad S(p) = 20 + 2p, \quad \gamma = 0.5.$$

- (a) Substitute demand, supply, and the adjustment parameter into the differential equation:

$$\frac{dp}{dt} = 0.5 [(80 - p) - (20 + 2p)].$$

- (b) Simplify inside the brackets:

$$(80 - p) - (20 + 2p) = 80 - p - 20 - 2p = 60 - 3p.$$

Therefore,

$$\frac{dp}{dt} = 0.5(60 - 3p).$$

Hence,

$$\frac{dp}{dt} = 30 - 1.5p.$$

- (c) A steady-state price satisfies

$$30 - 1.5p^* = 0.$$

Therefore,

$$1.5p^* = 30,$$

so

$$p^* = 20.$$

(d) The steady-state price is stable. If

$$p < 20,$$

then

$$30 - 1.5p > 0,$$

so price rises. If

$$p > 20,$$

then

$$30 - 1.5p < 0,$$

so price falls. Thus, price moves toward $p^* = 20$.

(e) Economically, the adjustment process says that price rises when demand exceeds supply and falls when supply exceeds demand. The parameter $\gamma = 0.5$ controls how quickly price adjusts. The steady state occurs where demand equals supply.

10. **Slightly harder problem.** Consider the Solow-style capital accumulation equation

$$k_{t+1} = sAk_t^{1/2} + (1 - \delta)k_t,$$

where

$$s = 0.4, \quad A = 10, \quad \delta = 0.2.$$

(a) Substitute the parameter values:

$$k_{t+1} = 0.4(10)k_t^{1/2} + (1 - 0.2)k_t.$$

Therefore,

$$k_{t+1} = 4k_t^{1/2} + 0.8k_t.$$

(b) A steady state satisfies

$$k^* = 4(k^*)^{1/2} + 0.8k^*.$$

Rearranging,

$$k^* - 0.8k^* = 4(k^*)^{1/2}.$$

Thus,

$$0.2k^* = 4(k^*)^{1/2}.$$

If $k^* > 0$, divide both sides by $(k^*)^{1/2}$:

$$0.2(k^*)^{1/2} = 4.$$

Therefore,

$$(k^*)^{1/2} = 20.$$

Squaring both sides,

$$k^* = 400.$$

There is also the steady state

$$k^* = 0,$$

since

$$0 = 4\sqrt{0} + 0.8(0).$$

The positive steady state is

$$k^* = 400.$$

(c) Suppose $k_0 = 100$. Then

$$k_1 = 4\sqrt{100} + 0.8(100).$$

Since

$$\sqrt{100} = 10,$$

we get

$$k_1 = 4(10) + 80 = 40 + 80 = 120.$$

(d) Suppose $k_0 = 900$. Then

$$k_1 = 4\sqrt{900} + 0.8(900).$$

Since

$$\sqrt{900} = 30,$$

we get

$$k_1 = 4(30) + 720 = 120 + 720 = 840.$$

(e) Capital moves toward the positive steady state $k^* = 400$. When

$$k_0 = 100 < 400,$$

we get

$$k_1 = 120 > 100,$$

so capital increases toward 400. When

$$k_0 = 900 > 400,$$

we get

$$k_1 = 840 < 900,$$

so capital decreases toward 400.

(f) Economically, the steady state is the level of capital at which new investment exactly offsets depreciation. At the positive steady state $k^* = 400$, the capital stock remains constant over time. If capital is below this level, investment is large enough to increase capital. If capital is above this level, depreciation dominates enough that capital decreases.